

LoRA Fine-Tuning Configuration

Parameter	Value
Base Model	Qwen2.5-7B (4-bit quantized)
LoRA Rank (r)	16
LoRA Alpha (α)	32
Target Modules	q, k, v, o, gate, up, down
Trainable Params	40.4M / 7.66B (0.53%)
Epochs	3
Batch Size	1 (× 8 grad accum)
Learning Rate	2×10^{-4} (cosine decay)
Dataset	84 train / 21 validation
Final Train Loss	0.764
Final Eval Loss	1.181
Training Time	~2 minutes (RTX 4060)